Resume Analysis and Performance Evaluation System

ABSTRACT

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1. Introduction

The Resume Analysis and Performance Evaluation System is a software-based application designed to automate the process of resume screening and candidate evaluation. In today's competitive job market, organizations receive a large number of applications for a single job opening, making manual review slow, inefficient, and error-prone. This system provides an intelligent approach to analyze resumes, match them against job requirements, and rank candidates, thereby improving the efficiency and quality of hiring decisions.

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2. Aim of the Project

The main aim of this project is to design and develop a system that can:

- Automate resume reading and information extraction
- Match candidate profiles with job descriptions
- Evaluate and rank candidates based on skills, experience, and qualifications
- Provide a fair, consistent, and unbiased evaluation process

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3. Problem Statement

Traditional recruitment methods rely heavily on manual resume screening, which:

- Consumes significant time and human effort
- Is inconsistent and prone to errors when handling large volumes of applications
- May introduce personal bias in candidate selection
- Does not provide feedback to candidates for improvement

This project addresses these challenges by automating the resume analysis and evaluation process.

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4. Methodology / Working

The system operates through the following steps:

• The user uploads a resume in PDF or DOCX format
• The system extracts textual content from the uploaded document
• Key information such as skills, education, experience, and certifications is identified
• The extracted data is compared with the given job description
• A scoring algorithm calculates a job-fit score using keyword-based matching techniques
• A skill gap analysis report is generated to highlight missing or weak areas
• Candidates are ranked based on their scores and results are displayed through a user-friendly interface

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5. Technologies Used

• Python (Flask) – Backend framework used for handling application logic and API requests
• HTML, CSS, JavaScript – Frontend technologies used to build the user interface
• MySQL – Database used to store resumes, job descriptions, and evaluation results
• pdfplumber – Library used for extracting text from PDF files
• python-docx – Library used for extracting text from DOCX files
• NLTK – Used for text preprocessing (tokenization, stopword removal)
• spaCy – Used for Named Entity Recognition (NER)
• TF-IDF (Scikit-learn) – Used for text vectorization and scoring

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6. Features of the System

• Resume upload and automated text extraction
• Extraction of skills, education, and experience
• Candidate scoring and ranking
• Skill gap identification
• Simple, efficient, and user-friendly interface

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8. Future Scope

• Integration of deep learning models (e.g., BERT) for better semantic matching
• Automated resume improvement suggestions
• Chatbot-based feedback system
• Expansion into a full-scale Applicant Tracking System (ATS)